

Aarushi Kotwani

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SUMMARY

Data Professional with 2+ years of experience developing ML models, NLP pipelines, and AI-driven analytics solutions across finance, sustainability, and research domains. Skilled in Python, SQL, and AWS cloud workflows; delivered production-ready LSTM forecasters, computer vision pipelines, and RAG-based LLM applications. Consistently converted analytical outputs into cost savings of \$80K+ and efficiency gains of 35% across cross-functional teams.

SKILLS

- **Languages:** Python, SQL, R
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, Keras, PyTorch
- **ML & Algorithms:** Regression, Classification, Clustering, Neural Networks, CNN, LSTM, XGBoost, ARIMA, Prophet, Forecasting
- **AI & GenAI:** LLMs, Prompt Engineering, RAG, LangChain, OpenAI API, Hugging Face Transformers, NLP, Computer Vision
- **MLOps & Deployment:** MLflow, AWS (S3, Glue, Redshift, Lambda), Docker, FastAPI, Git, GitHub Actions
- **Data Engineering:** ETL Pipelines, Spark, Feature Engineering, ChromaDB, Vector Databases, Data Cleaning
- **Analytics & Visualization:** Tableau, Power BI, A/B Testing, Hypothesis Testing, Statistical Modeling, Data Visualization, Excel

EXPERIENCE

Seattle University

Aug 2025 - Present

Research Assistant (Data Scientist)

Seattle, WA

- Engineered an end-to-end ML image analysis pipeline in Python (Scikit-learn, Pandas) to process 91+ biological samples, extracting 144-dimensional features from color, texture, and edge data - reducing manual review time by ~40%.
- Applied K-means clustering, PCA, and t-SNE for unsupervised pattern discovery, achieving a silhouette score of 0.38 and identifying 3+ distinct visual clusters across sample categories.
- Conducted statistical correlation analysis to separate imaging artifacts from biological variation, determining that 80% of observed differences stemmed from imaging conditions and improving experimental reproducibility.
- Promoted to independently lead data acquisition for 2 new study tracks; presented findings to faculty and senior researchers, reflecting expanded scope and direct research ownership.

Qualitest

Apr 2026 - May 2026

Data Scientist(Contract)

Kirkland, WA

- Curated and validated high-fidelity multimodal datasets for AI/ML model training, applying statistical quality checks in Python and Excel to maintain 98% data accuracy across high-volume annotation workflows.
- Built Python and SQL data-validation scripts to detect anomalies and enforce schema consistency across structured datasets, producing clean, ML-ready ground-truth data for Meta's AR/VR computer vision systems.
- Automated ETL ingestion and validation logging for model-training pipelines, and developed data visualization reports that delivered audit-ready, reproducible datasets supporting downstream ML development.

Amazon (Capstone Project)

Jan 2025 - Jun 2025

Data Scientist

Seattle, WA

- Architected an AWS data pipeline (S3, Glue, Lambda) to automate ingestion and validation of sustainability data across 500+ organizations, cutting data processing time by 35% and eliminating manual ETL effort.
- Developed regression and time-series forecasting models (ARIMA, Prophet) to uncover key sustainability performance drivers, enabling stakeholders to prioritize initiatives with the highest projected impact.
- Designed Tableau dashboards on SQL-transformed data to monitor KPIs and forecast trends in real time, enabling cross-functional teams to reduce decision lag by delivering insights 2x faster.

Seattle University

Jun 2024 - Jun 2025

Data Scientist Intern

Seattle, WA

- Analyzed 5+ years of financial spend data using Python and SQL, producing regression forecasting models that improved annual budget planning accuracy by 15% and surfaced \$80K+ in cost-saving opportunities.
- Deployed automated Tableau dashboards to track budget KPIs and variance trends across 10+ cost centers, cutting manual reporting workload by 15% and improving real-time visibility for finance leadership.
- Collaborated with finance leadership to translate model outputs into allocation strategies, validated findings with A/B and hypothesis testing in R, and delivered recommendations that improved allocation decision accuracy for quarterly budgeting.

PROJECTS

RAG-Based Document Chatbot

- Developed a RAG-based chatbot using LangChain, OpenAI API, and ChromaDB that enables users to query uploaded PDF documents with precise answers; implemented semantic chunking, vector embeddings, and prompt-engineered LLM workflows served via a Streamlit UI.
- Reduced average query resolution time by 60% compared to manual document search; achieved 100% context relevance across 50+ test queries with zero hallucinations by tuning retrieval parameters and refining prompts iteratively.

Impact of Environmental & Health Factors on Mental Health

- Processed 100K+ global health records (1990-2019) through a Python ETL pipeline and trained a Bidirectional LSTM model to forecast mental disorder prevalence, achieving $R^2 = 0.904$.
- Visualized trends across 20+ countries using Tableau and Excel dashboards, translating model outputs into public health strategy recommendations for policymakers.

Credit Card Fraud Detection System with MLOps Integration

- Developed a fraud detection system using Logistic Regression, Random Forest, and XGBoost to classify credit card transactions; deployed via Flask and Docker with LangChain/OpenAI API integration for natural language prediction explanations.
- Implemented real-time risk segmentation (high/medium/low) with fraud probability scoring, and built a CI/CD-ready pipeline using Scikit-learn, Pandas, and Docker for scalable model deployment.

EDUCATION

Seattle University <i>M.S., Data Science</i>	Sep 2023 - Jun 2025 <i>Seattle, WA</i>
Bundelkhand University <i>M.A., Economics</i>	Aug 2020 - Jun 2022 <i>Jhansi, UP, India</i>
Bundelkhand University <i>M.Com., Commerce</i>	Aug 2017 - Jun 2019 <i>Jhansi, UP, India</i>